



ECB Interest-Rate Pass-Through to Euro-Area Bank Lending Rates During the Post-2022 Tightening Cycle

Thesis submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in
Economics and Business Economics with a specialization in Economics

School of Business Economics

Vrije Universiteit Amsterdam

Author: Kevin Pinto

Supervisor: dr. Björn Brügemann

Second Reader: dr. Ana Figueiredo

26th of June, 2026

1. Introduction & Motivation

The period following the Covid-19 pandemic marked one of the most significant monetary policy shifts in the history of the euro area. After nearly a decade of exceptionally low and, at times, negative interest rates, inflation rose sharply between 2021 and 2023, reaching levels not observed since the introduction of the euro. In response, the European Central Bank (ECB) implemented one of the fastest monetary tightening cycles in its history, raising policy rates by a cumulative 450 basis points between July 2022 and September 2023. This abrupt transition from an environment of abundant liquidity and low borrowing costs to one characterized by rapidly rising interest rates renewed interest in a fundamental question in macroeconomics: how effectively do changes in ECB policy rates transmit to the broader economy?

According to New Keynesian monetary theory, changes in policy rates influence borrowing costs faced by firms and households, thereby affecting investment, consumption, and aggregate demand (Bernanke & Woodford, 1997; Woodford, 2003). In the euro area, where bank-based financing remains the dominant source of external funding for firms, the effectiveness of monetary policy depends on how quickly and to what extent banks pass policy-rate changes through to lending rates. If pass-through is delayed or incomplete, monetary policy becomes less effective in influencing credit conditions and inflation.

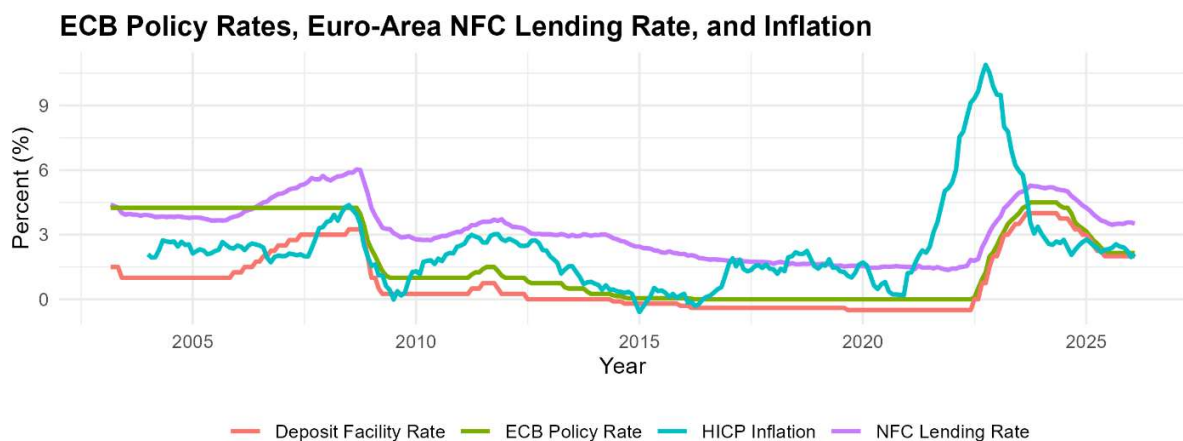


Figure 0: ECB policy rates, euro-area NFC lending rate, and HICP inflation.

Note: The figure plots monthly euro-area data.

The NFC lending rate captures the cost of borrowing for non-financial corporations.

The ECB policy rate refers to the Main Refinancing Operations (MRO) rate.

The Deposit Facility Rate (DFR) captures the interest rate paid on excess reserves.

Inflation is measured as the annual HICP inflation rate.

Source: European Central Bank (ECB) Data Portal.

Figure 0 illustrates the evolution of the ECB Main Refinancing Operations (MRO) rate, the Deposit Facility Rate (DFR), euro-area NFC lending rates, and annual HICP inflation between January 2003 and March 2026. The figure highlights the sharp increase in inflation after the pandemic, the rapid ECB tightening cycle beginning in July 2022, and the subsequent adjustment of NFC lending rates. These developments motivate the central question of whether the extent and timing of ECB interest-rate pass-through changed during the post-2022 tightening cycle, with the extent of pass-through referring to its overall magnitude, and the speed of pass-through referring to the timing of lending-rate adjustments. The

analysis focuses on these two dimensions of monetary transmission. This thesis contributes to the existing literature by extending the distributed-lag framework of Messner and Niepmann (2023) to euro-area NFC lending rates and by providing evidence on whether pass-through changed during the post-2022 ECB tightening cycle.

Zlobins, 2024 argues that pass-through to bank lending rates is gradual and incomplete. While (Beyer et al., 2024) indicate recent evidence suggesting that the magnitude and timing of pass-through vary across monetary regimes and financial products. The post-pandemic tightening cycle therefore raises the possibility that ECB interest-rate pass-through differed from earlier episodes. July 2022 provides a natural structural break for analyzing euro-area monetary transmission. The ECB's first policy-rate increase in more than a decade marked the end of the negative-interest-rate period and the beginning of one of the fastest tightening cycles in its history. This thesis therefore investigates the following research question:

Has ECB interest-rate pass-through to euro-area non-financial corporation (NFC) lending rates changed during the post-2022 tightening cycle?

The ECB's operational framework comprises three key policy rates: the Main Refinancing Operations (MRO) rate, the Deposit Facility Rate (DFR), and the Marginal Lending Facility (MLF) rate. This thesis uses the MRO rate as the baseline measure because of its traditional role as the ECB's primary policy rate and its widespread use in the interest-rate pass-through literature. The DFR is also considered as an alternative specification because, under the ECB's floor system with abundant excess reserves, it has become the key operational rate influencing short-term money-market conditions and banks' funding incentives (Brandão-Marques et al., 2024 & Hudepohl et al., 2024). The MLF is excluded because it primarily serves as the upper bound of the ECB's interest-rate corridor and plays a limited role in banks' funding decisions under conditions of abundant excess liquidity.

Using monthly euro-area data from January 2003 to March 2026, the thesis estimates a distributed-lag model inspired by Messner and Niepmann (2023). The baseline specification includes immediate and twelve-lagged ECB policy-rate changes and controls for immediate and lagged monthly HICP changes. The inclusion of HICP changes is drawn directly from the model used by Beyer et al., 2024. A post-2022 interaction model then tests whether the extent and speed of pass-through changed after the July 2022 tightening cycle began.

The results indicate that ECB policy-rate changes continue to exert a substantial influence on euro-area NFC lending rates. In the baseline model controlling for inflation, approximately 38% of an ECB policy-rate change is transmitted immediately to NFC lending rates. In the post-2022 interaction model, the immediate pass-through coefficient declines from approximately 48% before July 2022 to approximately 31% during the tightening cycle. Wald tests further indicate that pass-through remained statistically significant, while the positive interaction effect at the six-month horizon suggests that part of the adjustment shifted towards intermediate horizons. Overall, the findings suggest that the post-2022 tightening cycle suggests changes the timing of ECB pass-through more than its overall effectiveness.

2. Literature Review

The literature on monetary-policy pass-through examines how changes in central-bank policy rates are transmitted through financial markets and the banking sector to the borrowing costs faced by households and firms. While early studies focused primarily on the average extent of pass-through, more recent research has increasingly emphasized its timing, heterogeneity across financial products, and variation across monetary-policy regimes.

This chapter reviews the literature most relevant to the present study. It first discusses the theoretical foundations of the monetary transmission mechanism and interest-rate pass-through. It then reviews empirical evidence on the extent and speed of pass-through in the euro area, with particular attention to recent studies examining the post-pandemic tightening cycle. These studies provide the theoretical and empirical foundation for the distributed-lag models estimated in Chapter 4.

2.1 Monetary Policy Transmission

Theoretical models of monetary policy emphasize the central role of interest rates in shaping economic activity. Within the New Keynesian framework, changes in policy rates influence consumption and investment by altering real interest rates and, consequently, the intertemporal decisions of firms and households (Bernanke & Woodford, 1997). Higher policy rates increase firms' cost of capital and reduce the present value of future profits, lowering investment, while simultaneously encouraging household saving and reducing consumption (Woodford, 2003; Bernanke & Woodford, 1997).

This mechanism constitutes the interest-rate channel of monetary policy, through which central bank actions affect broader financial conditions and real economic activity (Taylor, 1993). In addition, modern monetary theory emphasizes the role of expectations, whereby anticipated future policy paths influence current economic decisions, implying that both current and expected future interest rates shape monetary transmission (Svensson, 2003).

Empirical evidence suggests that the transmission of policy rates to bank interest rates is gradual and incomplete. Early studies of the euro area show that changes in ECB policy rates are passed through to retail bank rates only gradually, with substantial heterogeneity across countries and financial products (de Bondt, 2005). More recent research further suggests that pass-through differs across lending and deposit products, reflecting differences in funding structures, repricing frequencies, and banking-sector characteristics (Messner & Niepmann, 2023 & Beyer et al., 2024). Recent evidence also suggests that the post-pandemic tightening cycle may have altered both the extent and speed of pass-through because of the unprecedented transition from negative interest rates to one of the fastest tightening cycles in ECB history (Messner & Niepmann, 2023; Beyer et al., 2024).

Together, these theoretical and empirical insights suggest that monetary-policy transmission through bank lending rates depends on institutional characteristics, market conditions, and the prevailing monetary-policy regime. This provides the theoretical foundation for examining whether ECB interest-rate pass-through to euro-area NFC lending rates changed during the post-2022 tightening cycle.

2.2 Interest Rate Pass-Through

A consistent finding in this literature is that monetary policy transmission operates with long and heterogeneous lags, implying that policy-rate changes affect financial markets, bank funding conditions, and retail interest rates gradually rather than instantaneously (Zlobins, 2024). In the euro area, ECB policy-rate changes are first reflected in market interest rates and bank funding costs before being passed through to lending and deposit rates offered to firms and households.

More recent evidence suggests that the monetary policy environment itself influences pass-through dynamics. Using a rolling error-correction framework, Michail and Louka (2025) show that the pass-through of market rates to euro-area bank interest rates evolved considerably over time. Their findings indicate that unconventional monetary policies strengthened pass-through for certain lending products, particularly NFC lending rates, while substantial heterogeneity persisted across countries and financial products. While (Ozturk, 2025) argues that ECB policy-rate changes are first transmitted to short-term interbank interest rates before affecting bank lending and deposit rates, indicating that money-market rates constitute an intermediate stage in the monetary transmission mechanism.

Early euro-area studies documented that pass-through is generally incomplete in the short run and varies substantially across products and countries. De Bondt (2005) finds that retail bank rates adjust gradually to changes in market interest rates, with lending and deposit products exhibiting different speeds and degrees of adjustment. This heterogeneity has been attributed to factors such as pricing rigidities, contractual frictions, differences in market competition, and bank-specific characteristics spanning the past two decades.

Recent evidence also suggests that the post-pandemic tightening cycle may have altered the broader monetary transmission mechanism. Zlobins (2024) estimates that conventional monetary policy shocks require up to 18 months to fully transmit to output and headline inflation in the euro area, highlighting the gradual nature of monetary-policy transmission. Consistent with this perspective, Beyer et al. (2024) and Messner and Niepmann (2023) employ dynamic distributed-lag models with contemporaneous and lagged policy-rate changes to estimate both the extent and speed of interest-rate pass-through. Together, these approaches form the empirical foundation for the distributed-lag framework developed in Chapter 3.

2.3 Time Variation and Structural Change

Changes in financial structure, monetary-policy regimes, and macroeconomic conditions may alter how policy-rate changes are transmitted to the economy (Zlobins, 2024; Curi & Murgia, 2023). Beyer et al. (2024) further suggest that the extent and timing of pass-through differ across monetary-policy regimes. Drawing on historical evidence from major macroeconomic events, including the introduction of the euro and the Global Financial Crisis, they argue that structural changes in the financial system help explain the heterogeneous transmission of policy rates to retail bank interest rates.

Using data on seven bank interest rates across 30 European countries, Beyer et al. (2024) argues that the post-pandemic tightening cycle was characterized by substantial heterogeneity across countries and financial products. Compared with earlier tightening episodes, pass-through was generally weaker and slower, although NFC lending rates exhibited relatively strong transmission. The authors further

attribute these differences to banking-sector characteristics, including liquidity conditions, market concentration, and lending opportunities.

Although previous studies document that monetary-policy transmission may vary across monetary regimes, relatively little evidence exists on whether the timing and cumulative magnitude of ECB pass-through to euro-area NFC lending rates changed following the July 2022 tightening cycle. This thesis addresses that gap by estimating a distributed-lag model that allows pass-through to differ before and after July 2022 through interaction terms. Rather than focusing solely on whether pass-through became stronger or weaker, the analysis examines whether the entire transmission profile shifted over time.

Overall, the literature suggests that interest-rate pass-through is gradual, incomplete, and potentially subject to structural change as financial conditions and monetary-policy regimes evolve. Building on these insights, this thesis tests whether the post-2022 tightening cycle altered not only the extent of ECB pass-through to euro-area NFC lending rates but also its timing. In particular, the empirical analysis evaluates whether immediate pass-through weakened, whether adjustments shifted toward later horizons, and whether these changes are jointly significant, thereby providing evidence on structural change in the monetary transmission mechanism.

3. Empirical Framework and Methodology

The literature reviewed in the previous section suggests that interest-rate pass-through is gradual, incomplete, and potentially subject to structural change across monetary regimes. To empirically examine these dimensions, this thesis employs a distributed lag framework inspired by Messner and Niepmann (2023), adapted to the analysis of euro-area NFC lending rates to answer the question:

Has ECB interest-rate pass-through to euro-area non-financial corporation (NFC) lending rates changed during the post-2022 tightening cycle?

3.1 Methodology

This thesis examines whether the extent and speed of ECB interest-rate pass-through to euro-area non-financial corporation (NFC) lending rates changed during the post-2022 monetary tightening cycle. To do so, it employs a distributed-lag framework that captures both the immediate and delayed effects of ECB policy-rate changes on lending rates.

The empirical strategy combines elements of the frameworks developed by Messner and Niepmann (2023) and Beyer et al. (2024). From Messner and Niepmann (2023), the analysis adopts a distributed-lag specification to estimate the timing and cumulative magnitude of pass-through. From Beyer et al. (2024), this analysis adopts the hypothesis that monetary transmission may differ across monetary-policy regimes by allowing pass-through dynamics to change following the ECB's July 2022 tightening cycle. To account for changing macroeconomic conditions and reduce the risk of omitted-variable bias, the preferred specifications additionally control for immediate and lagged monthly HICP inflation.

The empirical analysis proceeds in stages. It first estimates a baseline distributed-lag model, followed by a specification including inflation controls. The model is then extended with post-2022

interaction terms to test whether the timing and magnitude of pass-through changed during the recent tightening cycle. Finally, a series of robustness checks evaluates the sensitivity of the results to alternative ECB policy-rate measures and joint coefficient restrictions.

3.2 Empirical Framework

The empirical framework reflects the gradual nature of monetary-policy transmission through the banking sector. Changes in ECB policy rates are first transmitted to banks' funding conditions before being reflected in retail lending rates. Consequently, the analysis employs distributed-lag models that include immediate and lagged ECB policy-rate changes, allowing the timing and cumulative magnitude of pass-through to be estimated over a twelve-month horizon.

3.2.1 Baseline model

The baseline model is specified as follows:

$$\Delta LoanRate_t = \alpha + \sum_{k=0}^{12} \beta_k \Delta ECBRate_{t-k} + \varepsilon_t$$

where:

- $\Delta LoanRate_t$ is the monthly change in euro-area NFC lending rate;
- α captures the expected monthly change in NFC lending rate when all explanatory variables equal zero
- $\Delta ECBRate_{t-k}$ is the monthly change in the ECB policy rate lagged by k months
- β_k measures ECB interest-rate pass-through at lag k
- ε_t is the error term

The specification allows both immediate and delayed effects of ECB policy-rate changes on NFC lending to be estimated.

3.2.2 Baseline model controlling for inflation

In this model specification, we add Inflation as a control variable:

$$\Delta LoanRate_t = \alpha + \sum_{k=0}^{12} \beta_k \Delta ECBRate_{t-k} + \sum_{j=0}^{12} \gamma_j \Delta Inflation_{t-k} + \varepsilon_t$$

where:

- $\Delta LoanRate_t$ is the monthly change in euro-area NFC lending rate;
- α captures the expected monthly change in NFC lending rate when all explanatory variables equal zero;
- $\Delta ECBRate_{t-k}$ is monthly change in ECB policy rate in k -lags
- β_k measures ECB interest-rate pass-through at lag k
- $\Delta Inflation_t$ is monthly change in HICP inflation rate lagged by k months
- γ_k captures the relationship between monthly HICP inflation changes and changes in NFC lending rates at lag k .
- ε_t is the error term

This specification examines whether the relationship between ECB policy-rate changes and NFC lending-rate adjustments remains robust after accounting for inflationary conditions.

3.2.3 Post-2022 tightening model controlling for inflation

$$\Delta LoanRate_t = \alpha + \theta Post2022_t + \sum_{k=0}^{12} \beta_k \Delta ECBRate_{t-k} + \sum_{k=0}^{12} \delta_k (\Delta ECBRate_{t-k} \times Post2022_t) + \sum_{j=0}^{12} \gamma_j \Delta Inflation_{t-k} + \varepsilon_t$$

where:

- $\Delta LoanRate_t$ is the monthly change in euro-area NFC lending rate;
- α captures the expected monthly change in NFC lending rate when all explanatory variables equal zero;
- $Post2022_t$ is a dummy variable equal to 1 (from July 2022 onwards) and 0 otherwise;
- θ captures any shift in the average level of NFC lending-rate changes after July 2022;
- $\Delta ECBRate_{t-k}$ is monthly change in ECB policy rate lagged by k months;
- β_k measures ECB interest-rate pass-through at lag k before July 2022;
- δ_k measures the change in pass-through at lag k during the post-2022 tightening cycle;
- Therefore, the combined coefficients $\beta_k + \delta_k$ represents pass-through after July 2022;
- $\Delta Inflation_{t-k}$ is the monthly change in HICP inflation rate lagged by k months;
- γ_k is captures the effect of monthly HICP inflation changes at lag k ;
- ε_t is the error term.

The post-2022 interaction model controlling for inflation constitutes the preferred specification of this thesis. By simultaneously accounting for inflation dynamics and allowing pass-through to differ before and after July 2022, it provides the primary basis for answering the research question. Subsequent robustness checks assess whether the main findings are sensitive to alternative measures of the ECB's policy stance.

3.2.4 Test on joint-coefficients

In addition to individual coefficient analysis, this thesis conducts Wald tests to evaluate the joint significance of groups of explanatory variables. While individual coefficients provide information on the effects of specific variables at particular horizons, Wald tests assess whether groups of related coefficients are collectively different from zero. This provides complementary insight on whether inflation and changes in pass-through during the post-2022 tightening cycle contribute significantly to explaining changes in euro-area NFC lending rates.

First, a Wald test is conducted on the immediate and lagged inflation coefficients to assess whether inflation contributes significantly to explaining changes in NFC lending rates:

$$H_0: \gamma_0 = \gamma_1 = \dots = \gamma_{12} = 0$$

$$H_A: \text{At least one } \gamma_j \neq 0$$

Second, a Wald test is conducted on the immediate and lagged Post-2022 interaction coefficients to determine whether the pass-through mechanism changed following the onset of the ECB's tightening cycle:

$$H_0: \delta_0 = \delta_1 = \delta_2 = \dots = \delta_{12} = 0$$

$$H_A: \text{At least one } \delta_j \neq 0$$

To complement the coefficient estimates, two additional Wald tests evaluate whether the short-run (lags 1–3) and longer-run (lags 4–12) groups of baseline ECB policy-rate lag coefficients are jointly significant. These tests assess the statistical importance of the distributed lag structure rather than individual lag coefficients.

Formulas:

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_A: \text{At least one of } \beta_1, \beta_2, \beta_3 \neq 0$$

AND

$$H_0: \beta_4 = \beta_5 = \dots = \beta_{12} = 0$$

$$H_A: \text{At least one of } \beta_4, \dots, \beta_{12} \neq 0$$

3.2.5 Robustness checks

To assess the robustness of the main findings, this thesis estimates a series of alternative specifications. Robustness checks analyze whether the estimated extent and timing of ECB interest-rate pass-through depend on the choice of policy-rate measure or during one of the stages of the monetary transmission process explained in section 2.2.

The first robustness check replaces the Main Refinancing Operations (MRO) rate with the ECB Deposit Facility Rate (DFR). While the MRO rate has traditionally been regarded as the ECB's primary policy rate, the transition to a floor system with abundant excess reserves has increased the importance of the DFR as the key operational policy rate influencing short-term money-market conditions (Brandão-Marques & Ratnovski, 2024). During the recent tightening cycle, the DFR became the principal instrument through which ECB policy-rate changes were transmitted to money markets under conditions of abundant liquidity (Hudepohl et al., 2024). Consequently, estimating the model using the DFR provides a useful robustness check of whether the results are sensitive to the choice of ECB policy-rate measure.

The alternative specification is given by:

$$\Delta LoanRate_t = \alpha + \sum_{k=0}^{12} \beta_k \Delta DFR_{t-k} + \varepsilon_t$$

where:

- $\Delta LoanRate_t$ is the monthly change in the euro-area NFC lending rate;
- α captures the expected monthly change in DFR lending rate when all explanatory variables equal zero
- ΔDFR_{t-k} is the monthly change in the ECB Deposit Facility Rate at lag k ;

- β_k measures the Deposit Facility Rate pass-through to NFC lending rates at lag k ;
- ε_t is the error term.

The estimated DFR specification is subsequently compared with the baseline MRO model to assess whether the estimated extent and speed of pass-through are sensitive to the choice of ECB policy-rate measure.

The previous sections examined pass-through from ECB policy rates to NFC lending rates. However, ECB policy-rate changes are first transmitted to wholesale money-market rates before affecting retail bank lending rates (Beyer et al., 2024). To assess whether the observed delays in NFC lending rates originate in money markets or within the banking sector, the post-2022 interaction model is re-estimated using the monthly change in the three-month EURIBOR as the dependent variable. Because EURIBOR closely reflects banks' short-term funding costs, it provides a useful benchmark for distinguishing between wholesale and retail monetary-policy transmission. Comparing ECB pass-through to EURIBOR with pass-through to NFC lending rates therefore offers a robustness check on whether transmission delays arise primarily after policy-rate changes have been incorporated into money-market rates (Beyer et al., 2024; Ozturk, 2025).

The EURIBOR transmission specification is given by:

$$\Delta EURIBOR_t = \alpha + \sum_{k=0}^{12} \beta_k \Delta ECBRate_{t-k} + \sum_{k=0}^{12} \delta_k (\Delta ECBRate_{t-k} \times Post2022_t) + \varepsilon_t$$

where:

- $\Delta EURIBOR_t$ is the monthly change in the three-month EURIBOR rate;
- α represents the expected monthly change in three-month EURIBOR when all explanatory variables are equal to zero
- $\Delta ECBRate_{t-k}$ is the monthly change in the ECB policy rate lagged by k months;
- β_k measures the ECB policy-rate pass-through to the three-month EURIBOR at lag k before July 2022;
- $Post2022_t$ is a dummy variable equal to 1 from July 2022 onward and 0 otherwise;
- δ_k measures the change in ECB policy-rate pass-through to the three-month EURIBOR at lag k during the post-2022 tightening cycle
- Therefore, the combined coefficients $\beta_k + \delta_k$ represents ECB policy-rate pass-through to the three-month EURIBOR at lag k after July 2022;
- ε_t is the error term

Together, these robustness checks evaluate whether the main conclusions regarding the magnitude and timing of ECB interest-rate pass-through remain consistent across alternative measures of the ECB's monetary-policy stance and different stages of the transmission mechanism.

3.3 Data and Variable Construction

The empirical analysis uses monthly euro-area data covering the period from January 2003 to March 2026. The dependent variable is the euro-area NFC lending rate, which measures firms' borrowing costs within the bank lending channel. The main explanatory variable is the ECB MRO rate. Additional variables include the DFR, the three-month EURIBOR, and the HICP.

All series are converted to monthly frequency and merged by month. The NFC lending rate, HICP and EURIBOR are dated at monthly frequency. The ECB policy-rate series is first converted into a continuous daily series by carrying policy-rate values forward between ECB rate changes and is then aggregated to monthly averages. The DFR series is converted to monthly frequency using the final available value within each month.

The empirical models are estimated in monthly changes. The dependent variable is constructed as the monthly change in the NFC lending rate, while the main policy variable is the monthly change in the ECB policy rate. To capture delayed monetary transmission, twelve monthly lags of the policy-rate change are included. The same lag structure is also applied to the DFR in the alternative policy-rate specification and to EURIBOR in the money-market transmission exercise.

Inflation controls are constructed as monthly log changes in the HICP index, multiplied by 100. The regressions therefore use monthly HICP changes rather than annual inflation rates. In the inflation-controlled specifications, both the immediate HICP change and twelve monthly lags are included to account for macroeconomic conditions that may influence both ECB policy decisions and lending-rate adjustments.

The post-2022 dummy equals one from July 2022 onward and zero otherwise. Interaction terms are created by multiplying the immediate and lagged ECB policy-rate changes by this dummy, allowing the pass-through profile to differ before and after the start of the ECB tightening cycle.

First differencing and the inclusion of lags reduce the effective estimation sample. The baseline model contains 263 observations, while specifications including distributed HICP controls contain 251 observations. All models are estimated using ordinary least squares. Statistical inference is based on Newey–West heteroskedasticity- and autocorrelation-consistent standard errors with twelve lags.

4. Results

4.1 Baseline model

Table 1. Baseline Pass-Through Regression Results

	(1)
Constant	0.005 (0.006)
Change in ECB Policy Rate	0.415*** (0.050)
Change in ECB Policy Rate, t-1	0.339** (0.123)
Change in ECB Policy Rate, t-2	0.140+ (0.075)
Change in ECB Policy Rate, t-3	-0.031 (0.061)
Change in ECB Policy Rate, t-4	0.030 (0.077)
Change in ECB Policy Rate, t-5	-0.110+ (0.058)

	(1)
Change in ECB Policy Rate, t-6	-0.060 (0.082)
Change in ECB Policy Rate, t-7	0.030 (0.028)
Change in ECB Policy Rate, t-8	0.097 (0.059)
Change in ECB Policy Rate, t-9	0.013 (0.090)
Change in ECB Policy Rate, t-10	0.000 (0.041)
Change in ECB Policy Rate, t-11	0.002 (0.057)
Change in ECB Policy Rate, t-12	-0.046 (0.055)
Observations	263
R-squared	0.675
Adjusted R-squared	0.658
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Notes:	
Newey-West standard errors with 12 lags are reported in parentheses.	
The dependent variable is the monthly change in the euro-area NFC lending rate.	
Immediate and lagged ECB policy-rate changes are included (with lags up to twelve months).	

Interpretation Table 1

The results indicate that pass-through is strongest immediately following an ECB policy-rate change. Approximately 42% of a one percentage-point increase in the ECB policy rate is transmitted to NFC lending rates within the same month. Positive coefficients at the one- and two-month horizons further suggest that banks continue adjusting lending rates after the initial policy change, indicating that pass-through extends beyond the contemporaneous month.

Summing the contemporaneous and lagged coefficients yields a cumulative pass-through of approximately 0.82 over a twelve-month horizon. Thus, a one percentage-point increase in the ECB policy rate is associated with an eventual increase of roughly 0.82 percentage points in the euro-area NFC lending rate. This implies substantial, although not complete, pass-through from monetary policy to bank lending rates.

Although most coefficients beyond the second month are individually insignificant, this does not imply that pass-through ends after two months. Rather, the distributed-lag specification captures the cumulative adjustment of lending rates over time. The joint significance of the lag structure is formally evaluated using Wald tests in the following section.

Overall, the baseline specification provides evidence of strong ECB interest-rate pass-through to euro-area NFC lending rates and establishes the benchmark against which the subsequent model specifications are compared.

4.2 Baseline Pass-through controlling for inflation

This model examines whether the relationship between ECB policy-rate changes and NFC lending-rate adjustments remains robust after accounting for inflationary conditions. Results are displayed in *table 2* below:

Table 2. Pass-Through Regression with Distributed Inflation Controls

	(1)
Constant	-0.006 (0.009)
Change in ECB Policy Rate	0.381*** (0.052)
Change in ECB Policy Rate, t-1	0.355** (0.130)
Change in ECB Policy Rate, t-2	0.083 (0.078)
Change in ECB Policy Rate, t-3	-0.018 (0.058)
Change in ECB Policy Rate, t-4	0.067 (0.083)
Change in ECB Policy Rate, t-5	-0.135* (0.054)
Change in ECB Policy Rate, t-6	-0.057 (0.069)
Change in ECB Policy Rate, t-7	0.035 (0.035)
Change in ECB Policy Rate, t-8	0.083 (0.063)
Change in ECB Policy Rate, t-9	0.030 (0.086)
Change in ECB Policy Rate, t-10	0.005 (0.039)
Change in ECB Policy Rate, t-11	-0.013 (0.070)
Change in ECB Policy Rate, t-12	-0.047 (0.057)
Observations	251
R-squared	0.712
Adjusted R-squared	0.679

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Newey-West standard errors with 12 lags are reported in parentheses.

The dependent variable is the monthly change in the euro-area NFC lending rate.

The model includes contemporaneous and lagged ECB policy-rate changes up to twelve months.

The model also includes contemporaneous and twelve lagged monthly HICP controls.

(1)
HICP coefficients are omitted from the table for brevity.
The full specification including all HICP coefficients is reported in Appendix Table A.1.

Interpretation Table 2

The results indicate that the estimated pass-through is largely unchanged after accounting for inflation. Immediate pass-through remains strong, with approximately 38% of a one percentage-point increase in the ECB policy rate transmitted to NFC lending rates within the same month. The cumulative pass-through over the twelve-month horizon equals approximately 0.77, indicating substantial, although not complete, transmission from ECB policy rates to bank lending rates. These findings suggest that the baseline results are robust to controlling for inflation.

The inclusion of distributed inflation controls improves the model's explanatory power. The R^2 increases from 0.675 in the baseline specification to 0.712, while the adjusted R^2 rises from 0.658 to 0.679. This suggests that inflation contributes to explaining variation in NFC lending rates without materially altering the estimated pass-through from ECB policy rates.

To evaluate the relevance of the inflation controls, Table 3 reports a Wald test of the contemporaneous and lagged HICP coefficients. The null hypothesis that all inflation coefficients are jointly equal to zero is strongly rejected ($F = 5.299$, $p < 0.001$). This provides statistical evidence that inflation contributes significantly to explaining changes in NFC lending rates and supports its inclusion in the preferred specifications. At the same time, the stability of the ECB pass-through coefficients indicates that the estimated relationship between ECB policy rates and NFC lending rates is not driven solely by inflationary developments.

Table 3: Wald Test of Joint Significance of Monthly HICP Controls

Test	F-statistic	Df 1	Df 2	P-value
Joint significance of inflation lags (0-12 months)	5.333	13	224	2.45e-08

Notes:

The null hypothesis states that all inflation coefficients from t to $t-12$ are jointly equal to zero. Newey-West standard errors with 12 lags are used.

3.3 Results of Post-2022 pass-through regression controlling for inflation

The below regression results in *table 4* test whether the magnitude and timing of ECB policy rate pass-through to NFC lending rates changed during the recent tightening period.

Table 4: Post-2022 Pass-Through Regression Results controlling for inflation

	(1)
Constant	-0.012 (0.008)
Post-2022 Period	-0.038**

	(1)
	(0.013)
Change in ECB Policy Rate	0.483***
	(0.079)
Change in ECB Policy Rate, t-1	0.495+
	(0.275)
Change in ECB Policy Rate, t-2	-0.081
	(0.221)
Change in ECB Policy Rate, t-3	0.145
	(0.103)
Change in ECB Policy Rate, t-4	-0.032
	(0.094)
Change in ECB Policy Rate, t-5	-0.056
	(0.156)
Change in ECB Policy Rate, t-6	-0.195+
	(0.107)
Change in ECB Policy Rate, t-7	0.121
	(0.094)
Change in ECB Policy Rate, t-8	0.029
	(0.109)
Change in ECB Policy Rate, t-9	0.167
	(0.134)
Change in ECB Policy Rate, t-10	-0.012
	(0.058)
Change in ECB Policy Rate, t-11	-0.169+
	(0.086)
Change in ECB Policy Rate, t-12	0.111+
	(0.057)
Change in ECB Policy Rate $\times$ Post-2022	-0.177*
	(0.087)
Change in ECB Policy Rate, t-1 $\times$ Post-2022	-0.229
	(0.298)
Change in ECB Policy Rate, t-2 $\times$ Post-2022	0.164
	(0.231)
Change in ECB Policy Rate, t-3 $\times$ Post-2022	-0.227+
	(0.126)
Change in ECB Policy Rate, t-4 $\times$ Post-2022	0.078
	(0.115)
Change in ECB Policy Rate, t-5 $\times$ Post-2022	-0.045
	(0.161)
Change in ECB Policy Rate, t-6 $\times$ Post-2022	0.236*
	(0.115)
Change in ECB Policy Rate, t-7 $\times$ Post-2022	-0.104
	(0.126)
Change in ECB Policy Rate, t-8 $\times$ Post-2022	0.006
	(0.117)
Change in ECB Policy Rate, t-9 $\times$ Post-2022	-0.231+
	(0.135)
Change in ECB Policy Rate, t-10 $\times$ Post-2022	0.042
	(0.072)
Change in ECB Policy Rate, t-11 $\times$ Post-2022	0.216+

	(1)
Change in ECB Policy Rate, $t-12 \times \text{Post-2022}$	(0.119) -0.195** (0.075)
Observations	251
R-squared	0.746
Adjusted R-squared	0.697
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$	
Notes:	
Newey-West standard errors with 12 lags are reported in parentheses.	
The dependent variable is the monthly change in the euro-area NFC lending rate.	
Post-2022 equals 1 from July 2022 onward and 0 otherwise.	
The model includes contemporaneous and lagged ECB policy-rate changes up to twelve months.	
The model includes contemporaneous and twelve lagged monthly HICP controls.	
Inflation coefficients are omitted from the table for brevity and reported in Appendix Table A.2.	
Interaction terms test whether pass-through changed during the post-2022 tightening cycle.	

Table 4 Interpretation: Post-2022 Pass-Through Controlling for Inflation

The results indicate that ECB policy-rate pass-through remained substantial after July 2022, but its timing changed. The contemporaneous ECB policy-rate coefficient β_0 is estimated at 0.483 and is highly statistically significant, indicating that approximately 48% of a one percentage-point increase in the ECB policy rate was transmitted immediately to NFC lending rates prior to July 2022. The contemporaneous interaction coefficient δ_0 is estimated at - 0.177 and is statistically significant, suggesting that immediate pass-through declined to approximately 31% during the post-2022 tightening cycle ($0.483 - 0.177 = 0.306$). Thus, although ECB policy-rate changes continued to influence lending rates, a smaller proportion of the adjustment occurred within the same month.

The interaction coefficients further indicate that the timing of pass-through shifted following July 2022. While the immediate response became weaker, the positive interaction coefficient at the six-month horizon suggests stronger pass-through at intermediate horizons. Collectively, these results suggest that the post-2022 tightening cycle altered the timing of monetary-policy transmission rather than eliminating pass-through altogether.

The inclusion of distributed inflation controls does not materially alter the estimated pass-through profile observed in the baseline models. Immediate pass-through remains statistically significant and the contemporaneous interaction effect remains negative and significant, indicating that the observed change in transmission cannot be attributed solely to inflationary developments. The model explains a substantial share of the variation in monthly lending-rate changes ($R^2 = 0.746$; adjusted $R^2 = 0.697$), supporting its use as the preferred specification.

To assess whether the post-2022 interaction effects are jointly significant, Table 5 reports a Wald test of the contemporaneous and lagged interaction coefficients.

Table 5 Wald Test of Joint Significance of Post-2022 Interaction Effects

Test	F-statistic	Df 1	Df 2	P-value
Joint significance of Post-2022 interaction terms (t to t-12)	5.278	13	210	<1e-04

Notes:

The null hypothesis states that the contemporaneous and lagged Post-2022 interaction coefficients are jointly equal to zero.

The test evaluates whether ECB pass-through changed following the onset of the July 2022 tightening cycle. Newey-West standard errors with 12 lags are used.

Interpretation table 5

Table 5 reports the Wald Test of Joint Significance of the Post-2022 interaction effects. The null hypothesis that all contemporaneous and lagged interaction coefficients are jointly equal to zero is rejected ($F = 5.278$, $p < 0.001$). This indicates that the overall pattern of ECB interest-rate pass-through changed significantly following the onset of the July 2022 tightening cycle.

This result complements the coefficient estimates reported in Table 4. While the individual interaction coefficients identify the specific horizons at which pass-through changed, the Wald test provides evidence that these changes are jointly statistically significant. Together, the coefficient estimates and the Wald test provide strong evidence that the timing of ECB interest-rate pass-through to euro-area NFC lending rates differed between the pre-2022 and post-2022 periods.

The economic interpretation of these findings is discussed further in Chapter 6.

4.4 Wald Tests of Pass-Through Speed

Table 6 reports Wald tests of the joint significance of the short-run (lags 1–3) and longer-run (lags 4–12) ECB policy-rate coefficients in the preferred specification. While the individual regression coefficients identify the horizons at which pass-through is statistically significant, the Wald tests assess whether groups of lagged coefficients jointly contribute to explaining changes in euro-area NFC lending rates. Together, these tests provide direct evidence on the speed of ECB interest-rate pass-through.

Test	F-statistic	Df 1	Df 2	P-value
Short-run pass-through effects (lags 1-3)	9.178	3	210	<0.001
Longer-run pass-through effects (lags 4-12)	3.142	9	210	0.00142

Notes:

The table reports Wald tests of the joint significance of ECB policy-rate lag coefficients in the inflation-controlled Post-2022 interaction model, controlling for inflation.

The short-run test examines whether lags 1 to 3 are jointly equal to zero.

The longer-run test examines whether lags 4 to 12 are jointly equal to zero.

Newey-West standard errors with 12 lags are used.

Interpretation of Table 6

Table 6 reports Wald tests of the joint significance of the baseline ECB policy-rate lags in the short-run (lags 1–3) and longer-run (lags 4–12) in the preferred specification. Rather than assessing whether pass-through changed after July 2022, these tests evaluate whether groups of lagged ECB policy-rate changes jointly contribute to explaining monthly changes in euro-area NFC lending rates after controlling for inflation and allowing for post-2022 interaction effects.

The null hypothesis that the short-run lag coefficients (lags 1–3) are jointly equal to zero is strongly rejected ($F = 9.178$, $p < 0.001$). This indicates that the short-run lag structure is statistically important in explaining changes in NFC lending rates beyond the contemporaneous response.

The null hypothesis for the longer-run lag coefficients (lags 4–12) is likewise rejected ($F = 3.142$, $p = 0.001$). Although relatively few of these individual coefficients are statistically significant, they are jointly important, implying that the delayed effects of ECB policy-rate changes continue to contribute to lending-rate adjustments over time.

Together, these findings demonstrate that the distributed lag structure is statistically meaningful across both short and longer horizons. Consequently, the effects of ECB policy-rate changes cannot be evaluated solely on the basis of individual coefficients; considering the lag structure jointly provides a more complete picture of the transmission process. These results therefore reinforce the use of a distributed-lag specification in modelling ECB interest-rate pass-through.

These findings are consistent with the literature on euro-area interest-rate pass-through, which attributes gradual adjustment to pricing rigidities, contractual arrangements, funding structures, and competitive conditions (Beyer et al., 2024; Messner & Niepmann, 2023). Overall, the Wald tests provide additional evidence that ECB policy-rate transmission through the bank lending channel is dynamic, with lending-rate adjustments distributed over several months rather than occurring instantaneously. The joint significance of the lag structure indicates that banks continue adjusting lending rates well beyond the contemporaneous response.

4.5 Robustness Checks: Alternative ECB Policy Rate Measures

The preceding analysis measures monetary policy using the ECB Main Refinancing Operations (MRO) rate. As a robustness check, the baseline distributed-lag model is re-estimated using the ECB Deposit Facility Rate (DFR). Following years of unconventional monetary policy and abundant excess liquidity, the DFR has become an increasingly relevant indicator of the ECB's monetary-policy stance. Comparing the two specifications allows the analysis to assess whether the estimated pass-through profile depends on the choice of policy-rate measure.

The results are reported in Table 7.

Table 7. Deposit Facility Rate Pass-Through Regression Results

	(1)
Constant	-0.004 (0.004)
Change in Deposit Facility Rate	0.392*** (0.070)

	(1)
Change in Deposit Facility Rate, t-1	0.255*** (0.028)
Change in Deposit Facility Rate, t-2	0.165** (0.053)
Change in Deposit Facility Rate, t-3	0.037 (0.038)
Change in Deposit Facility Rate, t-4	-0.040 (0.033)
Change in Deposit Facility Rate, t-5	-0.071+ (0.037)
Change in Deposit Facility Rate, t-6	0.013 (0.032)
Change in Deposit Facility Rate, t-7	0.060+ (0.031)
Change in Deposit Facility Rate, t-8	0.102** (0.032)
Change in Deposit Facility Rate, t-9	-0.027 (0.031)
Change in Deposit Facility Rate, t-10	-0.032 (0.027)
Change in Deposit Facility Rate, t-11	-0.021 (0.019)
Change in Deposit Facility Rate, t-12	0.006 (0.030)
Observations	263
R-squared	0.717
Adjusted R-squared	0.702
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Notes:	
Newey-West standard errors with 12 lags are reported in parentheses.	
The dependent variable is the monthly change in the euro-area NFC lending rate.	
The model includes contemporaneous and lagged changes in the ECB Deposit Facility Rate up to twelve months.	

Interpretation of Table 7

Table 7 re-estimates the baseline distributed-lag model using changes in the ECB Deposit Facility Rate instead of the MRO rate. The results are highly comparable to those obtained in the baseline specification. The contemporaneous DFR coefficient remains positive and highly statistically significant, while the first two lag coefficients are likewise positive and significant, indicating that pass-through remains concentrated in the months immediately following an ECB policy-rate change.

The cumulative twelve-month pass-through equals approximately 0.84, compared with 0.82 in the baseline MRO specification. In addition, the DFR model explains a similar proportion of the variation in monthly lending-rate changes ($R^2 = 0.717$).

Overall, the similarity between the MRO and DFR specifications indicates that the estimated extent and speed of ECB interest-rate pass-through are robust to the choice of policy-rate measure.

Table 8 compares the two specifications directly.

Table 8. Comparison of MRO and DFR Pass-Through Estimates

Coefficient	MRO	DFR
Contemporaneous	0.415***	0.392***
Lag 1	0.339**	0.255***
Lag 2	0.14+	0.165**
Lag 3	-0.031	0.037
Lag 4	0.03	-0.04
Lag 5	-0.11+	-0.071+
Lag 6	-0.06	0.013
Lag 7	0.03	0.06+
Lag 8	0.097	0.102**
Lag 9	0.013	-0.027
Lag 10	0	-0.032
Lag 11	0.002	-0.021
Lag 12	-0.046	0.006
12-month cumulative Pass-Through	0.816	0.838
R-squared	0.675	0.717
Adjusted R-squared	0.658	0.702
Observations	263	263

Notes:

The table compares the baseline MRO specification with an alternative specification using the ECB Deposit Facility Rate. Both models include contemporaneous and lagged policy-rate changes up to twelve months.

Significance stars are based on Newey-West standard errors with 12 lags.

12-month cumulative pass-through is calculated as the sum of the contemporaneous coefficient and lag coefficients from t-1 to t-12.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Interpretation of Table 8

Table 8 directly compares the baseline specification using the ECB Main Refinancing Operations (MRO) rate with the alternative specification based on the Deposit Facility Rate (DFR). The comparison gives insight that the principal findings of this thesis are robust to the choice of ECB policy-rate measure.

Both specifications produce highly statistically significant contemporaneous pass-through coefficients and exhibit similar distributed-lag profiles. Moreover, the estimated twelve-month cumulative pass-through is nearly identical, amounting to 0.816 in the MRO specification and 0.838 in the DFR specification. These results indicate that the overall extent of ECB interest-rate pass-through is largely unaffected by whether monetary policy is measured using the MRO or the DFR.

The DFR specification achieves a modestly higher R^2 (0.717 versus 0.675), suggesting a slightly better fit to the data. However, the differences in explanatory power and the estimated pass-through

coefficients are relatively small. Consequently, there is little evidence that the choice of ECB policy-rate measure materially affects the estimated timing or extent of pass-through.

Overall, the comparison confirms that the central conclusions of this thesis are robust to alternative measures of the ECB's monetary-policy stance, strengthening confidence in the empirical findings presented throughout the analysis.

4.6 Money-Market Transmission: Evidence from EURIBOR

Table 9. ECB Policy Rate Pass-Through to EURIBOR

	(1)
Constant	0.016+ (0.009)
Change in ECB Policy Rate	0.801*** (0.082)
Change in ECB Policy Rate, t-1	0.447 (0.291)
Change in ECB Policy Rate, t-2	0.141 (0.188)
Change in ECB Policy Rate, t-3	0.086 (0.062)
Change in ECB Policy Rate, t-4	-0.211 (0.129)
Change in ECB Policy Rate, t-5	0.056 (0.096)
Change in ECB Policy Rate, t-6	-0.278** (0.098)
Change in ECB Policy Rate, t-7	0.164 (0.128)
Change in ECB Policy Rate, t-8	0.127 (0.087)
Change in ECB Policy Rate, t-9	0.152* (0.077)
Change in ECB Policy Rate, t-10	-0.036 (0.072)
Change in ECB Policy Rate, t-11	-0.062 (0.081)
Change in ECB Policy Rate, t-12	0.003 (0.077)
Change in ECB Policy Rate $\times$ Post-2022	-0.159 (0.115)
Change in ECB Policy Rate, t-1 $\times$ Post-2022	-0.158 (0.306)
Change in ECB Policy Rate, t-2 $\times$ Post-2022	-0.062 (0.204)
Change in ECB Policy Rate, t-3 $\times$ Post-2022	-0.228+ (0.135)
Change in ECB Policy Rate, t-4 $\times$ Post-2022	0.131 (0.136)

	(1)
Change in ECB Policy Rate, t-5 × Post-2022	-0.105 (0.119)
Change in ECB Policy Rate, t-6 × Post-2022	0.351*** (0.105)
Change in ECB Policy Rate, t-7 × Post-2022	-0.064 (0.133)
Change in ECB Policy Rate, t-8 × Post-2022	-0.113 (0.090)
Change in ECB Policy Rate, t-9 × Post-2022	-0.276** (0.088)
Change in ECB Policy Rate, t-10 × Post-2022	0.010 (0.084)
Change in ECB Policy Rate, t-11 × Post-2022	0.088 (0.091)
Change in ECB Policy Rate, t-12 × Post-2022	-0.042 (0.092)
Observations	263
R-squared	0.733
Adjusted R-squared	0.703
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Notes:	
Newey-West standard errors with 12 lags are reported in parentheses.	
The dependent variable is the monthly change in the 3-month EURIBOR rate.	
Post-2022 equals 1 from July 2022 onward and 0 otherwise.	
Interaction terms test whether ECB pass-through to money-market rates changed during the post-2022 tightening cycle.	

Table 9 examines whether ECB policy-rate pass-through to the three-month EURIBOR changed during the post-2022 tightening cycle. The results indicate that pass-through to money-market rates is both rapid and substantial. The immediate ECB policy-rate coefficient β_0 is estimated at 0.801 and is highly statistically significant, implying that approximately 80% of a one percentage-point increase in the ECB policy rate is transmitted to the three-month EURIBOR within the same month. This is considerably larger than the contemporaneous pass-through estimated for euro-area NFC lending rates.

The post-2022 interaction effects provide limited evidence that money-market transmission changed during the recent tightening cycle. Although the interaction effects at the six- and nine-month horizons are statistically significant, most interaction coefficients are not, indicating that the overall pass-through profile remained broadly stable following July 2022.

The model explains a substantial proportion of the variation in monthly EURIBOR changes ($R^2 = 0.733$), confirming the close relationship between ECB policy rates and short-term money-market rates.

Comparing these results with the NFC lending-rate regressions suggests that ECB policy-rate changes are transmitted more quickly to wholesale money markets than to retail lending rates. This finding is consistent with the view that the additional delays observed for NFC lending rates arise

primarily after policy-rate changes have been transmitted to money markets. While the present analysis cannot identify the precise mechanisms underlying these delays, the literature suggests that bank pricing behavior, funding structures, contractual arrangements, and competitive conditions may contribute to the slower adjustment of retail lending rates (Beyer et al., 2024; Messner & Niepmann, 2023).

5. Discussion

This thesis examined whether the extent and timing of ECB interest-rate pass-through to euro-area non-financial corporation (NFC) lending rates changed during the post-2022 tightening cycle. Using a series of distributed-lag specifications, the analysis evaluated both the magnitude of pass-through and the speed with which ECB policy-rate changes were transmitted to firms' borrowing costs.

The empirical results indicate that ECB policy-rate changes continue to exert a strong influence on euro-area NFC lending rates. The baseline specifications show that approximately 42% of a one percentage-point increase in the ECB policy rate is transmitted immediately to lending rates, while cumulative pass-through reaches approximately 80% over a twelve-month horizon. These findings confirm that the bank lending channel remains an important mechanism through which ECB monetary policy affects firms' borrowing costs and support the theoretical framework outlined in Chapter 2.

The results also show that pass-through is gradual rather than instantaneous. Although the immediate coefficient accounts for the largest share of the adjustment, the distributed-lag estimates and Wald tests demonstrate that lending rates continue to respond over subsequent months. The joint significance of both the short-run and longer-run lag structures indicates that monetary-policy transmission extends beyond the month of the policy-rate change, even though most adjustment occurs within the first few months. Together, these findings suggest that ECB interest-rate pass-through is characterized by rapid, but not immediate, adjustment.

The preferred post-2022 specification further indicates that the recent tightening cycle might altered the timing rather than the overall extent of monetary-policy transmission. Immediate pass-through became weaker following July 2022, while part of the adjustment shifted towards intermediate horizons. Moreover, the Wald test rejects the null hypothesis that the Post-2022 interaction effects are jointly equal to zero, providing statistical evidence that the pass-through profile changed following the onset of the ECB's tightening cycle.

These findings are broadly consistent with recent literature. Messner and Niepmann (2023) show that pass-through during the recent tightening cycle differed from earlier monetary-policy episodes, while Beyer et al. (2024) document substantial heterogeneity across banking products and countries. Similarly, Zlobins (2024) argues that monetary-policy transmission varies across monetary regimes. The results of this thesis support these studies by suggesting that the post-2022 tightening cycle primarily changed the timing of pass-through rather than its overall effectiveness.

Several economic mechanisms may explain this shift. Following years of unconventional monetary policy, euro-area banks entered the tightening cycle with unusually high levels of excess liquidity, reducing the immediate pressure to reprice loans after ECB rate increases. As funding conditions tightened and higher policy rates became more persistent, banks gradually passed these higher costs on to borrowers. Competitive considerations may also have contributed, as banks initially had incentives to

limit increases in lending rates to preserve loan demand before adjusting prices more fully as the tightening cycle progressed.

The robustness analysis further strengthens confidence in these findings. Replacing the Main Refinancing Operations (MRO) rate with the Deposit Facility Rate (DFR) produces highly comparable coefficient estimates and cumulative pass-through, indicating that the principal conclusions are robust to the choice of ECB policy-rate measure.

5.1 Policy Implications

The findings have several implications for monetary policy in the euro area. First, the results indicate that conventional interest-rate policy remains an effective instrument for influencing firms' borrowing costs despite years of unconventional monetary policy and abundant excess liquidity. Immediate pass-through remains economically large, while cumulative pass-through approaches 80%, suggesting that the bank lending channel continues to operate effectively.

Second, the results suggest that policymakers should take the timing of monetary-policy transmission into consideration. The post-2022 tightening cycle did not substantially weaken pass-through; rather, it delayed part of the adjustment. Consequently, assessments made shortly after a policy-rate change may underestimate its eventual effect on borrowing costs, investment, and economic activity.

Finally, the results highlight the importance of banking-sector conditions in shaping monetary transmission. Excess liquidity, funding costs, and competitive conditions appear to influence the speed with which banks adjust lending rates. Monitoring these factors alongside policy rates may therefore improve the ECB's assessment of monetary-policy transmission. The similarity of the MRO and DFR specifications further suggests that the effectiveness of monetary policy is not sensitive to the choice of operational policy-rate measure.

5.2 Limitations

Despite these contributions, several limitations should be acknowledged. First, the analysis relies on aggregate euro-area data, which mask heterogeneity across member states. Second, although the sample captures the complete post-2022 tightening cycle through March 2026, additional observations would allow future research to examine whether the observed changes persist as monetary policy normalizes. Third, the empirical framework is intentionally parsimonious and does not explicitly model bank-specific characteristics such as capitalization, liquidity, or market concentration, all of which may influence pass-through.

Future research could address these limitations by estimating country-level panel models that exploit cross-country variation in banking-sector characteristics. Such an approach would substantially increase the number of observations, facilitate direct comparison with Beyer et al. (2024), and provide a more detailed understanding of how the extent and speed of ECB interest-rate pass-through differ across euro-area banking systems.

6. Conclusion

This thesis examined whether ECB interest-rate pass-through to euro-area non-financial corporation (NFC) lending rates changed during the post-2022 monetary tightening cycle. Motivated by the unprecedented shift from a prolonged period of negative interest rates to one of the fastest tightening regimes within the euro area, the analysis estimated a series of distributed-lag regression models using monthly data from January 2003 to March 2026. The empirical framework focused on two key dimensions of monetary transmission identified in the literature: the extent of pass-through and the speed of adjustment.

The empirical results demonstrate that ECB policy-rate changes continue to exert a substantial influence on euro-area NFC lending rates. Across the baseline specifications, approximately forty percent of a one-percentage-point increase in the ECB policy rate is transmitted immediately to lending rates, while cumulative pass-through reaches roughly eighty percent over a twelve-month horizon. These findings confirm that the bank lending channel remains an important mechanism through which ECB monetary policy influences firms' borrowing costs.

The analysis further shows that pass-through is gradual rather than instantaneous. The distributed-lag estimates and Wald tests indicate that lending-rate adjustments continue beyond the month in which policy rates change, although most of the adjustment occurs within the first three months. Suggesting that ECB interest-rate transmission is characterized by rapid, but not immediate, adjustment.

The preferred post-2022 specification supports the hypothesis that the recent tightening cycle altered the transmission process. Immediate pass-through became weaker following July 2022, while part of the adjustment shifted towards intermediate horizons. Importantly, the joint significance of the interaction terms suggests these changes are statistically significant. Rather than indicating a breakdown of the monetary transmission mechanism, the results suggest that the timing of pass-through changed while its overall effectiveness remained substantial.

The robustness analyses further strengthen these conclusions. Using the Deposit Facility Rate instead of the Main Refinancing Operations rate produces highly comparable estimates, indicating that the principal findings are robust to alternative measures of the ECB's policy stance. Likewise, the EURIBOR specification shows that pass-through to money-market rates remains rapid and broadly unchanged following July 2022, suggesting that the additional delays observed for NFC lending rates arise primarily within the banking sector rather than in wholesale money markets.

Overall, this thesis contributes to the literature on euro-area monetary-policy transmission by extending recent distributed-lag pass-through frameworks to NFC lending rates and by suggesting how the post-2022 tightening cycle may have affected both the extent and timing of pass-through. While previous studies have primarily focused on deposit rates, money-market rates, or cross-country comparisons, this thesis shows that ECB policy-rate transmission to corporate lending rates remained substantial but became more gradual during the recent tightening cycle.

As additional post-pandemic data become available, future research should examine whether these changes represent a temporary response to an exceptional monetary-policy episode or a more permanent shift in the functioning of the euro-area bank lending channel. Extending the analysis to

country-level panel data would also provide valuable insights into how banking-sector characteristics shape monetary-policy transmission across the euro area.

Reference List

- Bernanke, B. S., & Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4), 27–48. <https://doi.org/10.1257/jep.9.4.27>
- Bernanke, B. S., & Woodford, M. (1997). *Inflation forecasts and monetary policy* (NBER Working Paper No. 6157). National Bureau of Economic Research. <https://doi.org/10.3386/w6157>
- Beyer, R., Chen, R., Li, C., Misch, F., Ozturk, E. O., & Ratnovski, L. (2024). Monetary policy pass-through to interest rates: Stylized facts from 30 European countries (IMF Working Paper No. WP/24/9). International Monetary Fund. <https://www.imf.org/en/Publications/WP/Issues/2024/01/26/Monetary-Policy-Pass-Through-to-Interest-Rates-Stylized-Facts-from-30-European-557792>
- Blanchard, O. and Bernanke, B., 2023.
What caused the U.S. pandemic-era inflation? NBER Working Paper, No. 29838.
- Brandao-Marques, L., & Ratnovski, L. (2024). The ECB's future monetary policy operational framework: Corridor or floor? (IMF Working Paper No. WP/24/56). International Monetary Fund.
- Curi, C. and Murgia, L.M., 2023.
Forecast targeting and financial stability: Evidence from the European Central Bank and Bank of England. *Finance Research Letters*, 51, 103486.
- de Bondt, G. J. (2005). Interest rate pass-through: Empirical results for the euro area. *German Economic Review*, 6(1), 37–78. <https://doi.org/10.1111/j.1468-0475.2005.00121.x>
- Drechsler, I., Savov, A., & Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4), 1819–1876. <https://doi.org/10.1093/qje/qjx019>
- European Central Bank. (n.d.). Cost of borrowing for non-financial corporations (new business), euro area (changing composition), monthly (Series: MIR.M.U2.B.A2I.AM.R.A.2240.EUR.N). ECB Data Portal. <https://data.ecb.europa.eu/data/datasets/MIR/MIR.M.U2.B.A2I.AM.R.A.2240.EUR.N>
- European Central Bank. (n.d.). Deposit facility – date of changes (raw data) – Level, Euro area, Daily – businessweek (Series: FM.B.U2.EUR.4F.KR.DFR.LEV). ECB Data Portal. <https://data.ecb.europa.eu/data/datasets/FM/FM.B.U2.EUR.4F.KR.DFR.LEV>
- European Central Bank. (n.d.). Euribor 3-month – Historical close, average of observations through period, euro area (changing composition), monthly (Series: FM.M.U2.EUR.RT.MM.EURIBOR3MD_.HSTA). ECB Data Portal. https://data.ecb.europa.eu/data/datasets/FM/FM.M.U2.EUR.RT.MM.EURIBOR3MD_.HSTA
- European Central Bank. (n.d.). Main refinancing operations – Fixed rate tenders (fixed rate) (date of changes) – Level, Euro area, Daily – businessweek (Series: FM.B.U2.EUR.4F.KR.MRR_FR.LEV). ECB Data Portal. https://data.ecb.europa.eu/data/datasets/FM/FM.B.U2.EUR.4F.KR.MRR_FR.LEV
- Eurostat. (n.d.). HICP – Monthly data (monthly rate of change) (Dataset: prc_hicp_minr). European Commission. https://ec.europa.eu/eurostat/databrowser/view/prc_hicp_minr/default/table?lang=en&category=prc.prc_hicp.prc_hicp_ecoicop2

Hudepohl, T., & Malderez, S. (Eds.). (2024). The use of the Eurosystem's monetary policy instruments and its monetary policy implementation framework in 2022 and 2023 (ECB Occasional Paper No. 355). European Central Bank.

Messner, T., & Niepmann, F. (2023). *What determines passthrough of policy rates to deposit rates in the euro area?* FED Notes. Board of Governors of the Federal Reserve System.
<https://doi.org/10.17016/2380-7172.3297>

Oet, M.V. and Lyytinen, K., 2017. Does financial stability matter to the Fed in setting US monetary policy? *Review of Finance*, 21(1), pp.389–432.

Ozturk, E. (2025). ECB monetary policy passthrough to bank interest rates during tightening: Republic of Kosovo (IMF Selected Issues Paper No. SIP/2025/004). International Monetary Fund.

Sims, C.A., 1992. Interpreting the macroeconomic time series facts: The effects of monetary policy. *European Economic Review*, 36(5), pp.975–1000.

Svensson, L.E.O., 2003. What is wrong with Taylor rules? Using judgment in monetary policy through targeting rules. *Journal of Economic Literature*, 41(2), pp.426–477.

Taylor, J.B., 1993. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39, pp.195–214.

Woodford, M. (2003). *Interest and prices: Foundations of a theory of monetary policy*. Princeton University Press.

Zhemelko, S.C. and Kočišová, K., 2026. Interest rate pass-through: A meta-analysis. (Working paper / journal pending publication details)

Zlobins, A., 2024. Monetary policy transmission in the euro area: Is this time different? (Working paper, Latvijas Banka)

Appendix

Table A.1: Pass-Through Regression with Distributed Inflation Controls

	(1)
Constant	-0.006 (0.009)
Change in ECB Policy Rate	0.381*** (0.052)
Change in ECB Policy Rate, t-1	0.355** (0.130)
Change in ECB Policy Rate, t-2	0.083 (0.078)
Change in ECB Policy Rate, t-3	-0.018 (0.058)
Change in ECB Policy Rate, t-4	0.067 (0.083)
Change in ECB Policy Rate, t-5	-0.135* (0.054)
Change in ECB Policy Rate, t-6	-0.057 (0.069)
Change in ECB Policy Rate, t-7	0.035 (0.035)
Change in ECB Policy Rate, t-8	0.083 (0.063)
Change in ECB Policy Rate, t-9	0.030 (0.086)
Change in ECB Policy Rate, t-10	0.005 (0.039)
Change in ECB Policy Rate, t-11	-0.013 (0.070)
Change in ECB Policy Rate, t-12	-0.047 (0.057)
Monthly HICP Change	0.027+ (0.016)
Monthly HICP Change, t-1	0.009 (0.009)
Monthly HICP Change, t-2	-0.023+ (0.012)
Monthly HICP Change, t-3	0.027 (0.016)
Monthly HICP Change, t-4	0.029*** (0.008)
Monthly HICP Change, t-5	-0.018 (0.012)
Monthly HICP Change, t-6	0.014 (0.012)
Monthly HICP Change, t-7	0.008 (0.009)
Monthly HICP Change, t-8	0.010 (0.008)

	(1)
Monthly HICP Change, t-9	-0.014 (0.016)
Monthly HICP Change, t-10	-0.005 (0.009)
Monthly HICP Change, t-11	-0.003 (0.010)
Monthly HICP Change, t-12	-0.002 (0.012)
Observations	251
R-squared	0.712
Adjusted R-squared	0.679
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	
Notes:	
Newey-West standard errors with 12 lags are reported in parentheses.	
The dependent variable is the monthly change in the euro-area NFC lending rate.	
The model includes contemporaneous and lagged ECB policy-rate changes up to twelve months.	
The model also includes contemporaneous and lagged monthly HICP changes up to twelve months.	
HICP is measured as an index with 2025 = 100.	

Table A2. Post-2022 Pass-Through Regression with Distributed Inflation Controls

	(1)
Constant	-0.012 (0.008)
Change in ECB Policy Rate	0.483*** (0.079)
Change in ECB Policy Rate, t-1	0.495+ (0.275)
Change in ECB Policy Rate, t-2	-0.081 (0.221)
Change in ECB Policy Rate, t-3	0.145 (0.103)
Change in ECB Policy Rate, t-4	-0.032 (0.094)
Change in ECB Policy Rate, t-5	-0.056 (0.156)
Change in ECB Policy Rate, t-6	-0.195+ (0.107)
Change in ECB Policy Rate, t-7	0.121 (0.094)
Change in ECB Policy Rate, t-8	0.029 (0.109)
Change in ECB Policy Rate, t-9	0.167 (0.134)
Change in ECB Policy Rate, t-10	-0.012

	(1)
	(0.058)
Change in ECB Policy Rate, t-11	-0.169+ (0.086)
Change in ECB Policy Rate, t-12	0.111+ (0.057)
Post-2022 Period	-0.038** (0.013)
Change in ECB Policy Rate $\times$ Post-2022	-0.177* (0.087)
Change in ECB Policy Rate, t-1 $\times$ Post-2022	-0.229 (0.298)
Change in ECB Policy Rate, t-2 $\times$ Post-2022	0.164 (0.231)
Change in ECB Policy Rate, t-3 $\times$ Post-2022	-0.227+ (0.126)
Change in ECB Policy Rate, t-4 $\times$ Post-2022	0.078 (0.115)
Change in ECB Policy Rate, t-5 $\times$ Post-2022	-0.045 (0.161)
Change in ECB Policy Rate, t-6 $\times$ Post-2022	0.236* (0.115)
Change in ECB Policy Rate, t-7 $\times$ Post-2022	-0.104 (0.126)
Change in ECB Policy Rate, t-8 $\times$ Post-2022	0.006 (0.117)
Change in ECB Policy Rate, t-9 $\times$ Post-2022	-0.231+ (0.135)
Change in ECB Policy Rate, t-10 $\times$ Post-2022	0.042 (0.072)
Change in ECB Policy Rate, t-11 $\times$ Post-2022	0.216+ (0.119)
Change in ECB Policy Rate, t-12 $\times$ Post-2022	-0.195** (0.075)
Monthly HICP Change	0.005 (0.014)
Monthly HICP Change, t-1	0.010 (0.009)
Monthly HICP Change, t-2	-0.021 (0.013)
Monthly HICP Change, t-3	0.029+ (0.016)
Monthly HICP Change, t-4	0.029** (0.009)
Monthly HICP Change, t-5	-0.014 (0.013)
Monthly HICP Change, t-6	0.025* (0.012)
Monthly HICP Change, t-7	0.018* (0.009)
Monthly HICP Change, t-8	0.023*

	(1)
	(0.011)
Monthly HICP Change, t-9	0.006
	(0.014)
Monthly HICP Change, t-10	0.012
	(0.009)
Monthly HICP Change, t-11	0.006
	(0.012)
Monthly HICP Change, t-12	0.026+
	(0.014)
Observations	251
R-squared	0.746
Adjusted R-squared	0.697

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes:

Newey-West standard errors with 12 lags are reported in parentheses.

The dependent variable is the monthly change in the euro-area NFC lending rate.

Post-2022 equals 1 from July 2022 onward and 0 otherwise.

The model includes contemporaneous and lagged ECB policy-rate changes up to twelve months.

The model also includes contemporaneous and lagged monthly HICP changes up to twelve months.

Interaction terms test whether pass-through changed during the post-2022 tightening cycle.